

MASTER_IMPLEMENTATION_PLAN

Master Implementation Plan: Path to 100% Completion

Project: HelixCode **Target Date:** February 2026 (10-Week Plan) **Goal:** 100% Feature Completion, Test Coverage, and Documentation.

Phase 1: Foundation & Critical Providers (Weeks 1-2)

Focus: Closing critical gaps in LLM support and stabilizing core modules.

1.1. Critical LLM Providers

- **Task 1.1.1:** Implement AnthropicProvider (internal/llm/anthropic_provider.go).
 - Support Claude 3.5 Sonnet, 3.7 Sonnet, Opus.
 - Implement Generate and GenerateStream.
- **Task 1.1.2:** Implement GeminiProvider (internal/llm/gemini_provider.go).
 - Support Gemini 2.0 Flash, 1.5 Pro.
 - Implement Native Multimodal support.
- **Task 1.1.3:** Update ProviderFactory to register new providers.

1.2. Core Test Coverage (The “Safety Net”)

- **Task 1.2.1:** internal/task Module.
 - Create task_test.go with testify/mock.
 - Cover CreateTask, UpdateTask, GetTask, DeleteTask.
 - **Goal:** >80% Coverage.
- **Task 1.2.2:** internal/auth Module.
 - Add integration tests for JWT flow.
 - Test session expiry and renewal.
 - **Goal:** >80% Coverage.

1.3. Configuration API Fixes

- **Task 1.3.1:** Implement RestoreConfig and ReloadConfig in internal/config/config_api.go.

- **Task 1.3.2:** Implement WebSocket handlers for live config updates.

Phase 2: Core Feature Completion (Weeks 3-4)

Focus: Enterprise readiness and completing the Memory system.

2.1. Enterprise Providers

- **Task 2.1.1:** Implement BedrockProvider (AWS).
- **Task 2.1.2:** Implement AzureProvider (Microsoft).
- **Task 2.1.3:** Add authentication strategies (IAM, Entra ID).

2.2. Memory System Completion

- **Task 2.2.1:** Complete WeaviateProvider.
 - Implement Store, Retrieve, Search methods (currently stubs).
- **Task 2.2.2:** Implement Mem0Provider and BaseAIProvider.

2.3. Tool System Expansion

- **Task 2.3.1:** Implement FileSystemTool (Read, Write, Search, Diff).
- **Task 2.3.2:** Implement ShellTool with sandbox/timeout.

Phase 3: Advanced Capabilities (Weeks 5-6)

Focus: Differentiating features and UI polish.

3.1. Advanced LLM Features

- **Task 3.1.1:** Implement **Extended Thinking** (Claude).
 - Add reasoning_effort param.
- **Task 3.1.2:** Implement **Prompt Caching** (Anthropic/DeepSeek).
 - Add cache control headers and logic.

3.2. Web & Editing Tools

- **Task 3.2.1:** Implement WebSearchTool and WebFetchTool.
- **Task 3.2.2:** Implement MultiFileEditTool (Atomic transactions).

3.3. UI Enhancements

- **Task 3.3.1:** Update Terminal UI (applications/terminal_ui).
 - Add “New Task” form.
 - Add Cognee toggle.

Phase 4: Enterprise & Security (Weeks 7-8)

Focus: Security, Performance, and remaining providers.

4.1. Remaining Providers

- **Task 4.1.1:** Implement VertexAIProvider and GroqProvider.
- **Task 4.1.2:** Implement MistralProvider.

4.2. Security & Performance

- **Task 4.2.1:** Conduct Security Audit (OWASP).
- **Task 4.2.2:** Run Performance Benchmarks with new providers.
- **Task 4.2.3:** Implement Rate Limiting and Cost Tracking.

Phase 5: Documentation, Education & Launch (Weeks 9-10)

Focus: User experience and educational resources.

5.1. Website Content

- **Task 5.1.1:** Create Landing Page (Features, Hero, CTA).
- **Task 5.1.2:** Create Documentation Portal (Hugo/Jekyll structure).
- **Task 5.1.3:** Create Blog/Updates section.

5.2. Video Courses

- **Task 5.2.1:** Develop scripts for “HelixCode Fundamentals”.
- **Task 5.2.2:** Develop scripts for “Advanced Agentic Workflows”.
- **Task 5.2.3:** Develop scripts for “Enterprise Integration”.

5.3. Final Polish

- **Task 5.3.1:** 100% Test Coverage Verification.
- **Task 5.3.2:** Final “Gold Master” Release Build.

Execution Strategy: 1. **No Broken Windows:** Every commit must pass CI. 2. **Test-First:** Write tests for new providers before implementation. 3. **Documentation-Driven:** Update docs as features are added.